Parametrization of a Tight-Binding Model for Ag Atomic Chains

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1 Introduction

Silver is a single-band ($5s^1$) metal: away from the d-bands, its electronic structure near the Fermi level is well described by a single, isolated s-orbital per atom. This makes Ag atomic chains an attractive testbed for tight-binding (TB) models, which in their simplest form also assume a single orbital per site. Unlike Au, where the d-bands sit close enough in energy to the s-band to require an explicit multi-orbital treatment, Ag retains a clean single-band character that lets a minimal TB Hamiltonian reproduce its essential physics.

A particularly useful feature of Ag chains is that they undergo a Peierls-like transition: under stretching, the uniform metallic chain becomes unstable to dimerization, opening a band gap and turning the chain into a semiconductor. This gives a single, simple model access to both a tunable metal-insulator transition and, once dimerized, a true gapped system in which excitonic physics (electron-hole pairs bound below the gap) can be studied. Combined with real-time electron dynamics, this opens the possibility of using Ehrenfest dynamics — where the nuclear positions respond to the electronic excitation — to study phenomena such as exciton self-trapping, in which an exciton locally distorts the lattice that helped create it.

Realizing this program requires a TB model whose parameters are quantitatively tied to Ag rather than chosen for generic qualitative behavior. This report documents that parametrization: Section 2 describes the reference DFT calculations, Section 3 introduces the TB model and its parameters, Section 4 details the optimization procedure used to fit those parameters, and Section 5 summarizes the result and its current limitations.

2 Reference DFT calculations

We performed a series of single-point density functional theory (DFT) calculations on one-dimensional Ag atomic chains using Quantum ESPRESSO, with the PBE functional, ultrasoft pseudopotentials, and a 32 k-point sampling. The calculations span lattice parameters from 5.2 to 6.2 Å and dimerization amplitudes d from 0 to 0.2, where d is defined in terms of the two alternating interatomic distances a_1 and a_2 as

$$d = \frac{a_1 - a_2}{a_1 + a_2}. \tag{1}$$

These calculations provide the reference potential energy surface (PES), band gap, and band dispersion used to parametrize the TB model described below. We note that PBE is known to systematically underestimate band gaps; since the TB model is fit directly to the PBE gap rather than to an experimental or higher-level value, the resulting parametrization should be understood as reproducing the PBE single-particle gap specifically. This is not a concern for the present purpose, since both the DFT reference and the TB model describe the same (single-particle) quantity.

Ag was chosen over Au for this purpose because the Au d-bands sit close in energy to the relevant s-band, breaking the single-band picture that the TB model assumes. The Ag s-band is comparatively well isolated, making it a much better match to a simple one-orbital chain model.

3 Tight-binding chain model

The TB model uses a single s-orbital per site, with a hopping integral $\beta(r)$ that depends on the interatomic distance r:

$$\beta(r) = \beta(r_{eq}) + A\,(r - r_{eq})^q, \tag{2}$$

where r_{eq} is the equilibrium interatomic distance, and the coefficient A and exponent q control how strongly the hopping splits into two distinct values as the chain dimerizes. The equilibrium hopping integral was found to be $\beta(r_{eq}) = -0.668653$ eV, while A and q were fit to reproduce the DFT band gap opening under dimerization, giving $A = 0.007215487659$ and $q = 1$. Figure 1 compares the resulting band gap, as a function of lattice length and dimerization, between DFT and the TB model: the TB model reproduces the DFT result that the band gap depends on dimerization alone, independent of the underlying lattice length.

The value of $\beta(r_{eq})$ itself is fixed by a separate criterion: matching the band dispersion (bandwidth) of the DFT valence and conduction bands. This match is necessarily approximate, since the TB bands of this model are symmetric about the gap by construction — the conduction and valence bands are mirror images of one another — whereas the DFT bands of Ag are not: although similar in shape, the two bands have different depths (bandwidths). We therefore choose $\beta(r_{eq})$ so that the TB bandwidth equals the average of the DFT valence and conduction bandwidths, where each bandwidth is taken as the difference between the largest and smallest band energy across k. This choice is necessarily a compromise rather than an exact match to either DFT band individually.

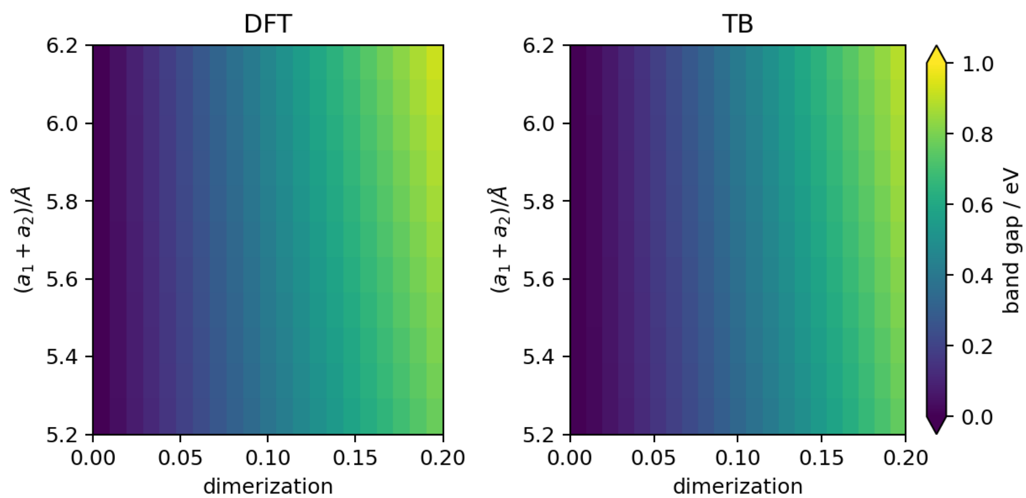

Figure 1: Band gap as a function of chain lattice length and dimerization d. Left: DFT results. Right: TB results for a periodic chain of 100 atoms, using $A = 0.007215487659$ and $q = 1$.

In addition to the hopping term, the atoms interact through a pairwise repulsive potential,

$$V(r) = -\beta(r_{eq})\,B\left(\frac{r_{eq}}{r}\right)^p, \tag{3}$$

whose role is to stabilize the equilibrium spacing r_{eq} against the attractive contribution of the hopping term. Since $\beta(r_{eq}) < 0$ in our convention, $V(r) > 0$ for all $r > 0$, as required for a repulsive term. (We note that since $\beta(r_{eq})$ and B both appear only as a product here and are both optimized, one of the two is in principle redundant; we kept both for consistency with the original parametrization scheme.) The exponent p was determined separately from $\beta(r_{eq})$, r_{eq}, and B: we repeated the full optimization described in Section 4 for fixed integer values of p and selected the value that minimized the resulting energy difference. This difference decreases monotonically from $p = 12$ to $p = 15$ and increases again slightly at $p = 16$, identifying $p = 15$ as the optimal exponent.

Together, the hopping and repulsive terms allow the chain to dimerize spontaneously under stretching, and provide the energetics needed for subsequent Ehrenfest dynamics simulations, where the nuclei move in response to electronic excitation. The next section describes how the remaining free parameters of this model — $\beta(r_{eq})$, r_{eq}, and B — were determined for Ag.

4 Parameter optimization

The goal of the optimization is to find the set of TB parameters that best reproduces the DFT potential energy surface. At each candidate parameter set, we evaluate the TB energy at the same geometries sampled by the DFT calculations and compute the summed squared deviation ΔE between the two surfaces. Points more than 0.1 eV above the DFT energy minimum are excluded from this sum. This cutoff is not merely a convenience: it is chosen to match the range of configurations expected to be sampled during subsequent Ehrenfest dynamics simulations, since those are the trajectories this parametrization is intended to support, and including the higher-energy points disproportionately degrades the fit quality within that range. Even so, the optimized PES remains in reasonable agreement with DFT at higher energies as well (Figure 2). We note that this agreement has not been systematically verified outside the fitted range, and we recommend that Ehrenfest trajectories be monitored to confirm they remain within the configurations this parametrization was validated against.

Parameters are optimized by Monte Carlo sampling over $\beta(r_{eq})$, r_{eq}, and B. At each step, all three parameters are perturbed by a random amount, $\delta\beta(r_{eq}) = \mathrm{RN} \times 0.0001$ eV, $\delta r_{eq} = \mathrm{RN} \times 0.0001$ Å, and $\delta B = \mathrm{RN} \times 0.001$, where RN is a random number drawn uniformly from $[0, 1]$. The move is accepted if it reduces ΔE, and otherwise accepted with probability

$$P_{acc} = \exp\left(-\frac{\Delta E_{i+1} - \Delta E_i}{kT}\right), \tag{4}$$

where kT is not a physical temperature but an annealing parameter controlling the acceptance of uphill moves. The annealing schedule consists of three stages of 1000 steps each: $kT = 0.0005$, then $kT = 0.0001$, and finally $kT = 0$ (i.e. a greedy descent) for the last 1000 steps. We also tested a steepest-descent approach, but found it more sensitive to the initial guess; with a sufficiently good starting point it likely converges to comparable parameters, but the Monte Carlo approach proved more robust in practice.

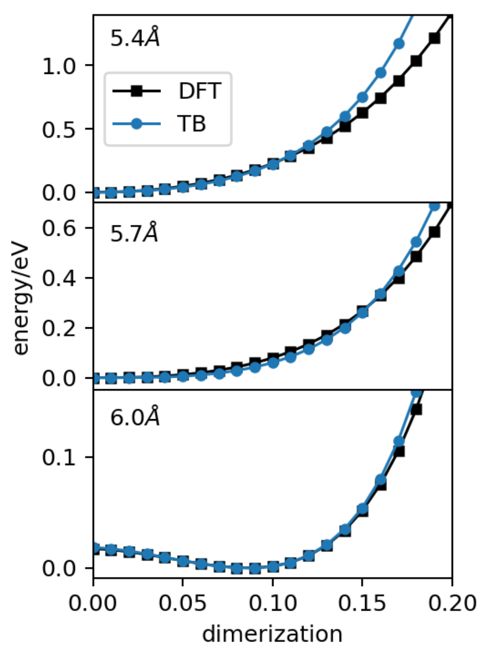

Figure 2: Energy per atom pair as a function of dimerization d, for three chain lattice lengths, comparing TB (100-atom chains) and DFT results. Optimized parameters: $r_{eq} = 2.60482$ Å, $B = 0.231122$, $p = 15$.

At convergence, the mean absolute energy difference between the TB and DFT potential energy surfaces, averaged over the 84 points used in the optimization, is 0.0026 eV.

Figure 2 shows the optimized PES as a function of dimerization for three chain lattice lengths. For lattice lengths beyond approximately 6 Å, the dimerized configuration becomes energetically favored over the uniform chain.

5 Conclusion

We have parametrized a single-orbital TB model for Ag atomic chains by fitting to DFT reference calculations, matching the band gap and average band dispersion to set the hopping parameters, and matching the potential energy surface near its minimum to set the repulsive interaction. The resulting model reproduces the DFT band gap as a function of dimerization across the tested range of lattice lengths, with a mean absolute energy deviation of 0.0026 eV relative to DFT over the fitted configurations.

This parametrization is deliberately restricted to the range of geometries expected during Ehrenfest dynamics simulations, rather than to the full range explored in the DFT scan. The model should accordingly be treated as validated within that range; trajectories that stray significantly outside it should be flagged for review rather than assumed to remain accurate. With this caveat, the model provides the energetic and electronic structure foundation needed for the next stage of this project: real-time electron dynamics on Ag chains, with the eventual goal of studying exciton formation and, through Ehrenfest dynamics, the possibility of exciton self-trapping.